

9.8 From (9.40) we have:

$$\mathbb{E}_{\mathbf{Z}}[\ln p(\mathbf{X}, \mathbf{Z} | \boldsymbol{\mu}, \boldsymbol{\Sigma}, \boldsymbol{\pi})] = \sum_{n=1}^N \sum_{k=1}^K \gamma(z_{nk}) \{\ln \pi_k + \ln \mathcal{N}(\mathbf{x}_n | \boldsymbol{\mu}_k, \boldsymbol{\Sigma}_k)\}$$

Taking the partial derivative w.r.t $\boldsymbol{\mu}_k$ while keeping responsibilities fixed, we get:

$$\begin{aligned} & \sum_{n=1}^N \gamma(z_{nk}) \frac{\partial}{\partial \boldsymbol{\mu}_k} \{\ln \pi_k + \ln \mathcal{N}(\mathbf{x}_n | \boldsymbol{\mu}_k, \boldsymbol{\Sigma}_k)\} \\ &= \sum_{n=1}^N \gamma(z_{nk}) \left\{ \mathbf{0} + \frac{\partial \ln \mathcal{N}(\mathbf{x}_n | \boldsymbol{\mu}_k, \boldsymbol{\Sigma}_k)}{\partial \boldsymbol{\mu}_k} \right\} \\ &= \sum_{n=1}^N \gamma(z_{nk}) \left\{ \frac{\partial \ln \mathcal{N}(\mathbf{x}_n | \boldsymbol{\mu}_k, \boldsymbol{\Sigma}_k)}{\partial \boldsymbol{\mu}_k} \right\} \end{aligned}$$

Using (2.118) and eliminating constants, we can see this becomes:

$$\begin{aligned} &= \sum_{n=1}^N \gamma(z_{nk}) \frac{\partial}{\partial \boldsymbol{\mu}_k} \left\{ -\frac{1}{2} (\mathbf{x}_n - \boldsymbol{\mu}_k)^T \boldsymbol{\Sigma}_k^{-1} (\mathbf{x}_n - \boldsymbol{\mu}_k) \right\} \\ &= \sum_{n=1}^N \gamma(z_{nk}) \{ \boldsymbol{\Sigma}_k^{-1} (\mathbf{x}_n - \boldsymbol{\mu}_k) \} \end{aligned}$$

We can pull $\boldsymbol{\Sigma}_k^{-1}$ out of the summation and set this to $\mathbf{0}$, giving us:

$$\begin{aligned} & \boldsymbol{\Sigma}_k^{-1} \sum_{n=1}^N \gamma(z_{nk}) \{ (\mathbf{x}_n - \boldsymbol{\mu}_k) \} = \mathbf{0} \\ & \implies \sum_{n=1}^N \gamma(z_{nk}) \{ (\mathbf{x}_n - \boldsymbol{\mu}_k) \} = \mathbf{0} \\ & \implies \sum_{n=1}^N \gamma(z_{nk}) \boldsymbol{\mu}_k = \sum_{n=1}^N \gamma(z_{nk}) \mathbf{x}_n \\ & \implies \boldsymbol{\mu}_k \sum_{n=1}^N \gamma(z_{nk}) = \sum_{n=1}^N \gamma(z_{nk}) \mathbf{x}_n \end{aligned}$$

$$\implies \boldsymbol{\mu}_k N_k = \sum_{n=1}^N \gamma(z_{nk}) \mathbf{x}_n$$

$$\implies \boldsymbol{\mu}_k = \frac{1}{N_k} \sum_{n=1}^N \gamma(z_{nk}) \mathbf{x}_n$$

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